along the shortest possible paths—the “easiest possible paths” or the “paths of least resistance.” If the space is warped such paths will be curved. And so, although the rubber membrane-bowling ball model provides a good visual analogy of how an object such as the sun warps the space around it and thereby influences the motion of other bodies, the physical mechanism by which these distortions occur is totally different. The former appeals to our intuition about gravity in the traditional Newtonian framework, whereas the latter expresses a reformulation of gravity in terms of curved space.

A second shortcoming of the analogy stems from the rubber membrane’s being two-dimensional. In reality, although harder to visualize, the sun (and all other massive objects) actually warps the three-dimensional space surrounding it. Figure 3.6 is a rough attempt to depict this; all of the space surrounding the sun—“below,” “on the sides,” on “top”—suffers the same kind of distortion, and Figure 3.6 schematically shows a partial sampling. A body, like the earth, travels through the three-dimensional warped spatial environment caused by the sun’s presence. You may find this figure troubling—why doesn’t the earth slam into the “vertical part” of curved space in the image? Bear in mind, though, that space, unlike the rubber membrane, is not a solid barrier. Instead, the warped grids in the image are but a couple of thin slices through the full three-dimensional warped space in which you, the earth, and everything else are immersed fully and move freely. Perhaps you find that this only makes the problem seem worse: Why don’t we feel space if we are immersed within its fabric? But we do. We feet gravity, and space is the medium by which the gravitational force is communicated. As the eminent physicist John Wheeler has often said in describing gravity, “mass grips space by telling it how to curve, space grips mass by telling it how to move.”8

A third, related shortcoming of the analogy is that we have suppressed the time dimension. We have done this for visual clarity because, notwithstanding the declaration of special relativity that we should think of the time dimension on par with the three familiar spatial dimensions, it is significantly harder to “see” time. But, as illustrated by the example of the Tornado ride, acceleration—and hence gravity—warps both space and time. (In fact, the mathematics of general relativity shows that in the case of a relatively slow-moving